

ELITE IGCSE ACADEMY

Student Past Paper Solutions

Jan 2021 P1H Worked Solutions

Jan 2021 | P1H

29 questions ■ question image followed by worked solution

PREPARED BY

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PAST PAPER QUESTION**Q#: 1****Marks: 4****TOPIC: Ratio Problem Solving**

Jan 2021 P1H - Jan 2021 | P1H

- 1** Pieter owns a currency conversion shop.

Last Monday, Pieter changed a total of 20 160 rand into a number of different currencies.

He changed $\frac{3}{10}$ of the 20 160 rand into euros.

He changed the rest of the rands into dollars, rupees and francs in the ratios 9 : 5 : 2

Pieter changed more rands into dollars than he changed into francs.

Work out how many more.

..... rand

(Total for Question 1 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 1****Marks: 4****TOPIC: Ratio Problem Solving**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. of the money is changed** $\frac{3}{10}$ of the money is changed to euros, so the rest is

$$\frac{7}{10} \times 20160 = 14112$$

2. Split the ratio

This rest is divided in the ratio

$$\text{dollars} : \text{rupees} : \text{francs} = 9 : 5 : 2$$

3. There are

There are

$$9 + 5 + 2 = 16 \text{ parts}$$

4. Split the ratio

The difference between dollars and francs is

$$9 - 2 = 7 \text{ parts}$$

5. Split the ratio

So the difference is

$$\frac{7}{16} \times 14112 = 6174$$

FINAL ANSWER

6174 rand more is changed into dollars than francs.

PAST PAPER QUESTION**Q#: 2****Marks: 5****TOPIC: Statistics Toolkit**

Jan 2021 P1H - Jan 2021 | P1H

- 2 The table gives information about the speeds, in kilometres per hour, of 80 motorbikes as each pass under a bridge.

Speed (s kilometres per hour)	Frequency
$40 < s \leq 50$	10
$50 < s \leq 60$	16
$60 < s \leq 70$	19
$70 < s \leq 80$	23
$80 < s \leq 90$	12

- (a) Write down the modal class.

.....
(1)

- (b) Work out an estimate for the mean speed of the motorbikes as they pass under the bridge.
Give your answer correct to 3 significant figures.

..... kilometres per hour

(4)

(Total for Question 2 is 5 marks)

PAST PAPER SOLUTION**Q#: 2** **Marks: 5****TOPIC: Statistics Toolkit**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. (a) The greatest frequency is**

(a) The greatest frequency is 23, so the modal class is

$$70 < s \leq 80$$

2. (b) Use midpoints

(b) Use midpoints 45, 55, 65, 75, 85:

$$\frac{45(10) + 55(16) + 65(19) + 75(23) + 85(12)}{80} = 66.375$$

3. To 3 significant figures

To 3 significant figures:

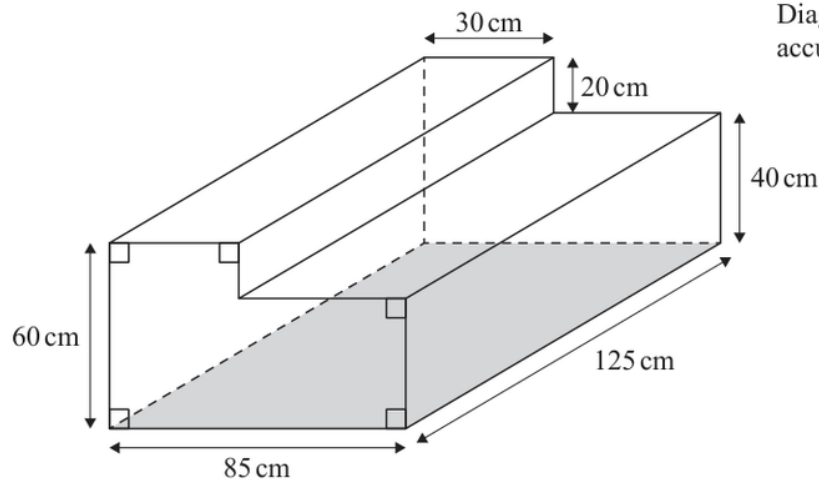
$$66.4$$

FINAL ANSWER(a) $70 < s \leq 80$, (b) 66.4 km/h.

PAST PAPER QUESTION**Q#: 3****Marks: 4****TOPIC: Volume & Surface Area**

Jan 2021 P1H - Jan 2021 | P1H

- 3 The diagram shows a container for water in the shape of a prism.



The rectangular base of the prism, shown shaded in the diagram, is horizontal.
The container is completely full of water.

Tuah is going to use a pump to empty the water from the container so that the volume of water in the container decreases at a constant rate.

The pump starts to empty water from the container at 10 30 and at 12 00 the water level in the container has dropped by 20 cm.

Find the time at which all the water has been pumped out of the container.

(Total for Question 3 is 4 marks)

PAST PAPER SOLUTION**Q#: 3****Marks: 4****TOPIC: Volume & Surface Area**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The cross section is an**

The cross-section is an L-shape.

2. Area of the lower rectangle

Area of the lower rectangle:

$$85 \times 40 = 3400$$

3. Area of the upper rectangle

Area of the upper rectangle:

$$30 \times 20 = 600$$

4. Calculate area

So the cross-sectional area is

$$3400 + 600 = 4000 \text{ cm}^2$$

5. The full volume is

The full volume is

$$4000 \times 125 = 500000 \text{ cm}^3$$

6. Calculate area

In the first 1.5 hours, the water level drops by 20 cm. This is the top rectangular part:

$$30 \times 20 \times 125 = 75000 \text{ cm}^3$$

7. Calculate area

So the pumping rate is

$$\frac{75000}{1.5} = 50000 \text{ cm}^3/\text{hour}$$

8. Time to empty the whole

Time to empty the whole container:

$$\frac{500000}{50000} = 10 \text{ hours}$$

9. Calculate area

Starting at 10:30, ten hours later is

20:30

FINAL ANSWER

20:30

EA

PAST PAPER QUESTION**Q#: 4****Marks: 2****TOPIC: Set Notation & Venn Diagrams**

Jan 2021 P1H - Jan 2021 | P1H

4 $\mathcal{E} = \{20, 21, 22, 23, 24, 25, 26, 27, 28, 29\}$

$$A = \{\text{odd numbers}\}$$

$$B = \{\text{multiples of 3}\}$$

List the members of the set

(i) $A \cap B$

.....
(1)

(ii) $A \cup B$

.....
(1)

.....
(Total for Question 4 is 2 marks)

PAST PAPER SOLUTION**Q#: 4****Marks: 2****TOPIC: Set Notation & Venn Diagrams**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate value**

$$\mathcal{E} = \{20, 21, 22, 23, 24, 25, 26, 27, 28, 29\}$$

2. The odd numbers are

The odd numbers are

$$A = \{21, 23, 25, 27, 29\}$$

3. The multiples of are

The multiples of 3 are

$$B = \{21, 24, 27\}$$

4. The numbers in both sets

The numbers in both sets are

$$A \cap B = \{21, 27\}$$

5. The numbers in either set

The numbers in either set are

$$A \cup B = \{21, 23, 24, 25, 27, 29\}$$

FINAL ANSWER(i) $\{21, 27\}$, (ii) $\{21, 23, 24, 25, 27, 29\}$

PAST PAPER QUESTION**Q#: 5****Marks: 5****TOPIC: Solving Linear Equations**

Jan 2021 P1H - Jan 2021 | P1H

5 (a) Factorise fully $15y^4 + 20uy^3$
(2)(b) Solve $4 - 3x = \frac{5 - 8x}{4}$

Show clear algebraic working.

 $x =$
(3)

(Total for Question 5 is 5 marks)

EA

PAST PAPER SOLUTION**Q#: 5****Marks: 5****TOPIC: Solving Linear Equations**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. (a)**

(a)

$$15y^4 + 20uy^3 = 5y^3(3y + 4u)$$

2. (b)

(b)

$$4 - 3x = \frac{5 - 8x}{4}$$

$$16 - 12x = 5 - 8x$$

$$11 = 4x$$

FINAL ANSWER(a) $5y^3(3y + 4u)$, (b) $x = \frac{11}{4}$

PAST PAPER QUESTION

Q#: 6

Marks: 2

TOPIC: Powers, Roots & Standard Form

Jan 2021 P1H - Jan 2021 | P1H

6 (a) Write 2840000000 in standard form.

.....
(1)

(b) Write 2.5×10^{-4} as an ordinary number.

.....
(1)

.....
(Total for Question 6 is 2 marks)

EA

PAST PAPER SOLUTION**Q#: 6****Marks: 2****TOPIC: Powers, Roots & Standard Form**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Convert standard form**

$$2840000000 = 2.84 \times 10^9$$

2. Convert standard form

Also,

$$2.5 \times 10^{-4} = 0.00025$$

FINAL ANSWER(a) 2.84×10^9 , (b) 0.00025

EA

PAST PAPER QUESTION**Q#: 7** **Marks: 6****TOPIC: Compound Interest & Depreciation**

Jan 2021 P1H - Jan 2021 | P1H

- 7 Chen invests 40 000 yuan in a fixed-term bond for 3 years.

The fixed-term bond pays compound interest at a rate of 3.5% each year.

- (a) Work out the value of Chen's investment at the end of 3 years.
Give your answer to the nearest yuan.

..... yuan
(3)

Wang invested P yuan.

The value of his investment decreased by 6.5% each year.

At the end of the first year, the value of Wang's investment was 30 481 yuan.

- (b) Work out the value of P .

$P =$
(3)

(Total for Question 7 is 6 marks)

PAST PAPER SOLUTION**Q#: 7****Marks: 6****TOPIC: Compound Interest & Depreciation**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use compound interest**

For part (a),

$$40000(1.035)^3 = 44348.715$$

2. Use compound interest

Correct to the nearest yuan,

$$44348.715 \approx 44349$$

3. Use compound interestFor part (b), after one year the value is 93.5% of P .

$$0.935P = 30481$$

$$P = \frac{30481}{0.935} = 32600$$

FINAL ANSWER(a) 44349 yuan, (b) $P = 32600$

PAST PAPER QUESTION**Q#: 8** **Marks: 3****TOPIC: Circles, Arcs & Sectors**

Jan 2021 P1H - Jan 2021 | P1H

- 8 The region, shown shaded in the diagram, is a path.

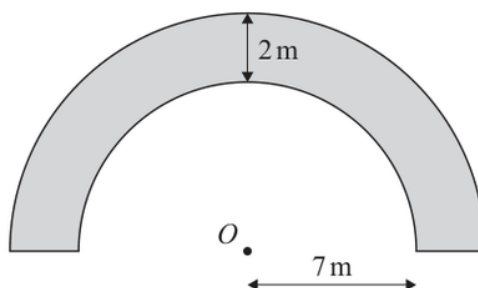


Diagram **NOT**
accurately drawn

The boundary of the path is formed by two semicircles, with the same centre O , and two straight lines.

The inner semicircle has a radius of 7 metres.

The path has a width of 2 metres.

Work out the perimeter of the path.

Give your answer correct to one decimal place.

..... m

(Total for Question 8 is 3 marks)

PAST PAPER SOLUTION**Q#: 8****Marks: 3****TOPIC: Circles, Arcs & Sectors**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The inner radius is m**

The inner radius is 7 m.

2. Calculate value

The path is 2 m wide, so the outer radius is

$$7 + 2 = 9 \text{ m}$$

3. Perimeter of the path

Perimeter of the path:

inner semicircle + outer semicircle + 2 straight ends

$$= 7\pi + 9\pi + 2 + 2$$

$$= 16\pi + 4$$

$$= 54.265 \dots$$

FINAL ANSWER

54.3 m.

PAST PAPER QUESTION**Q#: 9****Marks: 5****TOPIC: Factorising**

Jan 2021 P1H - Jan 2021 | P1H

9 (a) Simplify $(2x^3y^5)^4$
(2)(b) (i) Factorise $x^2 + 5x - 36$
(2)(ii) Hence, solve $x^2 + 5x - 36 = 0$
(1)

(Total for Question 9 is 5 marks)

EA

PAST PAPER SOLUTION**Q#: 9****Marks: 5****TOPIC: Factorising**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. (a)**

(a)

$$(2x^3y^5)^4 = 16x^{12}y^{20}$$

2. (b)(i)

(b)(i)

$$x^2 + 5x - 36 = (x + 9)(x - 4)$$

3. Factorise expression(b)(ii) $x = -9$ or $x = 4$.**FINAL ANSWER**(a) $16x^{12}y^{20}$, (b)(i) $(x + 9)(x - 4)$, (b)(ii) $x = -9$ or $x = 4$

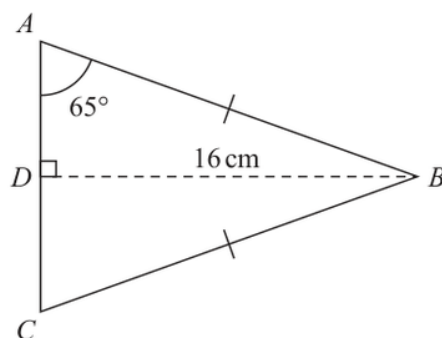
PAST PAPER QUESTION

Q#: 10

Marks: 4

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

Jan 2021 P1H - Jan 2021 | P1H

10 Here is isosceles triangle ABC .Diagram **NOT**
accurately drawn D is the midpoint of AC and $DB = 16$ cm.Angle $DAB = 65^\circ$ Work out the perimeter of triangle ABC .

Give your answer correct to one decimal place.

..... cm

(Total for Question 10 is 4 marks)

PAST PAPER SOLUTION**Q#: 10****Marks: 4****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Find the gradient**

Since ABC is isosceles and D is the midpoint of AC , BD is perpendicular to AC .

2. Use trigonometry

In right triangle ABD ,

$$BD = 16, \quad \angle DAB = 65^\circ$$

$$\sin 65^\circ = \frac{16}{AB}$$

$$AB = \frac{16}{\sin 65^\circ}$$

3. Use trigonometry

Also,

$$\tan 65^\circ = \frac{16}{AD}$$

$$AD = \frac{16}{\tan 65^\circ}$$

4. The perimeter is

The perimeter is

$$AB + BC + AC = 2AB + 2AD$$

$$= 2 \left(\frac{16}{\sin 65^\circ} \right) + 2 \left(\frac{16}{\tan 65^\circ} \right)$$

$$= 50.2299 \dots$$

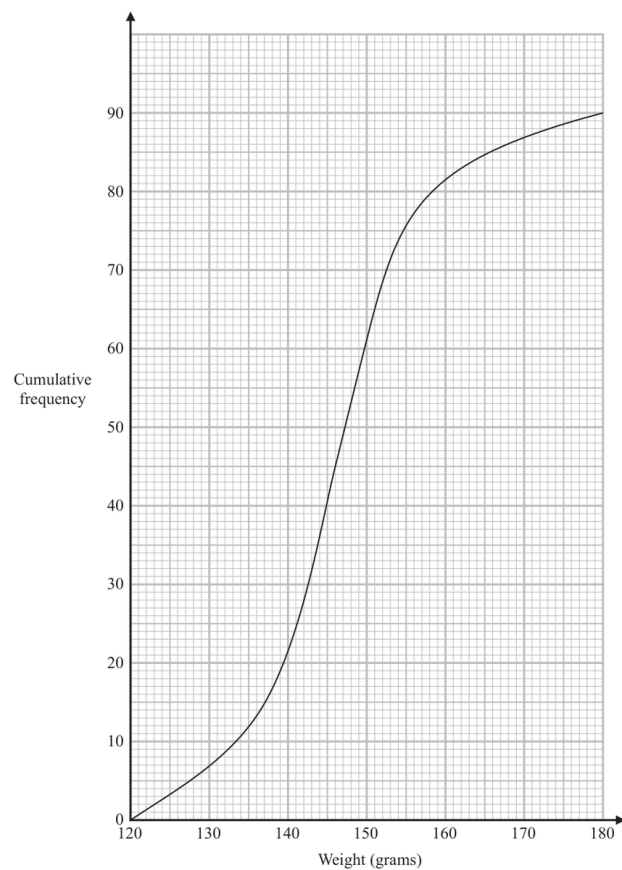
FINAL ANSWER

50.2 cm.

PAST PAPER QUESTION**Q#: 11** **Marks: 5****TOPIC: Cumulative Frequency Diagrams**

Jan 2021 P1H - Jan 2021 | P1H

- 11 The cumulative frequency graph gives information about the weights, in grams, of 90 bags of sweets.



- (a) Find an estimate for the median of the weights of these bags of sweets.

..... grams
(2)

Roberto sells the bags of sweets to raise money for charity.
Bags with a weight greater than d grams are labelled large bags and sold for 3.75 euros each bag.

The total amount of money he receives by selling all the large bags is 93.75 euros.

- (b) Find the value of d .

$d =$
(3)

(Total for Question 11 is 5 marks)

PAST PAPER SOLUTION**Q#: 11****Marks: 5****TOPIC: Cumulative Frequency Diagrams**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use cumulative frequency**

There are 90 bags, so the median is at cumulative frequency 45.

2. Use cumulative frequency

From the graph,

$$\text{median} \approx 146 \text{ g}$$

3. Large bags sold

Large bags sold:

$$\frac{93.75}{3.75} = 25$$

4. Use cumulative frequency

So 25 bags have weight greater than d .

5. That means

That means

$$90 - 25 = 65$$

6. Use cumulative frequency

bags have weight d grams or less.

7. Use cumulative frequency

From the graph, cumulative frequency 65 gives

$$d \approx 150$$

FINAL ANSWER

median about 146 g, and $d \approx 150$.

PAST PAPER QUESTION**Q#: 12****Marks: 6****TOPIC: Expanding Brackets**

Jan 2021 P1H - Jan 2021 | P1H

12 (a) Express $\frac{4}{x-2} - \frac{3}{x+1}$ as a single fraction.

Give your answer in its simplest form.

.....
(3)

(b) Expand and simplify $2x(x-5)(x-3)$

.....
(3)

(Total for Question 12 is 6 marks)

EA

PAST PAPER SOLUTION**Q#: 12****Marks: 6****TOPIC: Expanding Brackets**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. (a)**

(a)

$$\begin{aligned}\frac{4}{x-2} - \frac{3}{x+1} &= \frac{4(x+1) - 3(x-2)}{(x-2)(x+1)} \\ &= \frac{4x+4-3x+6}{(x-2)(x+1)} = \frac{x+10}{(x-2)(x+1)}\end{aligned}$$

2. (b)

(b)

$$\begin{aligned}2x(x-5)(x-3) &= 2x(x^2 - 8x + 15) \\ &= 2x^3 - 16x^2 + 30x\end{aligned}$$

FINAL ANSWER(a) $\frac{x+10}{(x-2)(x+1)}$, (b) $2x^3 - 16x^2 + 30x$

PAST PAPER QUESTION**Q#: 13****Marks: 3****TOPIC: Coordinate Geometry**

Jan 2021 P1H - Jan 2021 | P1H

- 13** Point A has coordinates $(5, 8)$
Point B has coordinates $(9, -4)$

(a) Work out the gradient of AB .

.....
(2)

The straight line L has equation $y = -4x + 5$

(b) Write down the gradient of a straight line that is perpendicular to L .

.....
(1)

.....
(Total for Question 13 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 13****Marks: 3**TOPIC: **Coordinate Geometry**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Find the gradient**(a) The gradient of AB , where $A(5, 8)$ and $B(9, -4)$, is

$$\frac{-4 - 8}{9 - 5} = \frac{-12}{4} = -3$$

2. Find the gradient(b) A line perpendicular to a line with gradient -3 has gradient

$$\frac{1}{3}$$

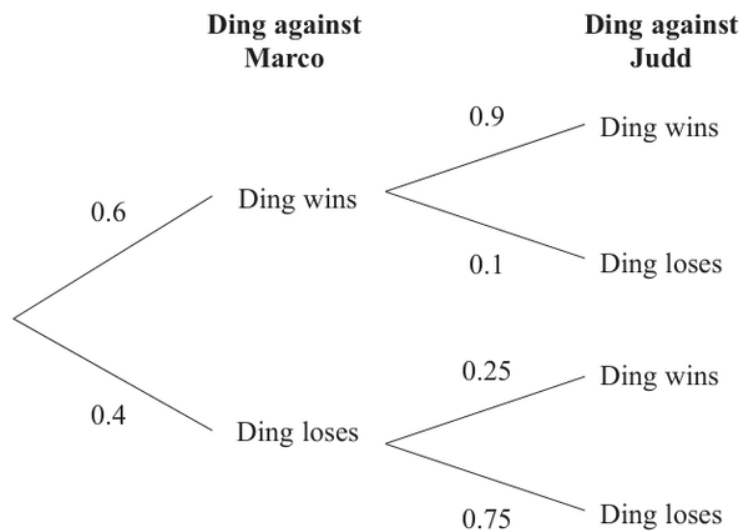
FINAL ANSWERGradient of $AB = -3$, perpendicular gradient $\frac{1}{3}$.

PAST PAPER QUESTION**Q#: 14****Marks: 5****TOPIC: Probability Diagrams - Venn & Tree Diagrams**

Jan 2021 P1H - Jan 2021 | P1H

- 14** Ding is going to play one game of snooker against each of two of his friends, Marco and Judd.

The probability tree diagram gives information about the probabilities that Ding will win or lose each of these two games.



- (a) Work out the probability that Ding will win both games.

.....
(2)

- (b) Work out the probability that Ding will win exactly one of the games.

.....
(3)

(Total for Question 14 is 5 marks)

PAST PAPER SOLUTION**Q#: 14****Marks: 5****TOPIC: Probability Diagrams - Venn & Tree Diagrams**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Winning both frames**

Winning both frames:

$$0.6 \times 0.9 = 0.54$$

2. Winning exactly one frame

Winning exactly one frame:

$$0.6 \times 0.1 + 0.4 \times 0.25$$

$$= 0.06 + 0.10 = 0.16$$

FINAL ANSWER

0.54, 0.16.

PAST PAPER QUESTION**Q#: 15****Marks: 3****TOPIC: Rounding, Estimation & Bounds**

Jan 2021 P1H - Jan 2021 | P1H

15

$$a = \frac{v - u}{t}$$

 $v = 9.6$ correct to 1 decimal place $u = 3.8$ correct to 1 decimal place $t = 1.84$ correct to 2 decimal placesCalculate the upper bound for the value of a .

Give your answer as a decimal correct to 2 decimal places.

Show your working clearly.

(Total for Question 15 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 15****Marks: 3****TOPIC: Rounding, Estimation & Bounds**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Estimate the value**

$$a = \frac{v - u}{t}$$

2. Find upper bound

For the upper bound, use the upper bound of v , the lower bound of u , and the lower bound of t .

$$v < 9.65, \quad u \geq 3.75, \quad t \geq 1.835$$

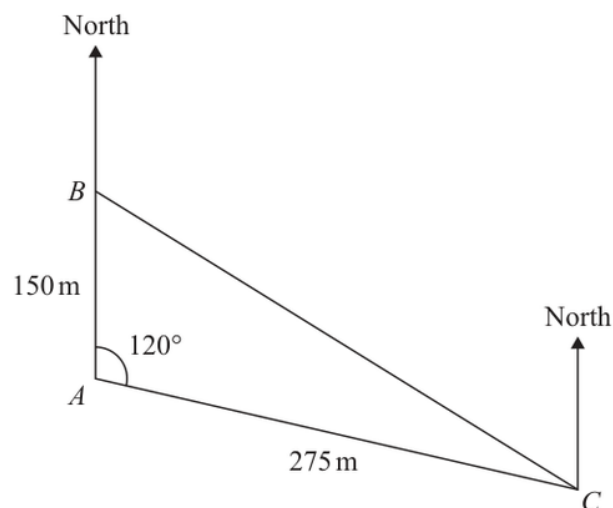
$$a_{\text{upper}} = \frac{9.65 - 3.75}{1.835} = 3.2152 \dots$$

FINAL ANSWER**3.22**

EAA

PAST PAPER QUESTION**Q#: 16****Marks: 5****TOPIC: Bearings, Scale Drawing & Constructions**

Jan 2021 P1H - Jan 2021 | P1H

16 The diagram shows the positions of three ships, A , B and C .Diagram **NOT**
accurately drawnShip B is due north of ship A .The bearing of ship C from ship A is 120° .Calculate the bearing of ship C from ship B .

Give your answer correct to the nearest degree.

(Total for Question 16 is 5 marks)

PAST PAPER SOLUTION**Q#: 16****Marks: 5****TOPIC: Bearings, Scale Drawing & Constructions**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use bearings**

From A , ship C is on a bearing of 120° , and $AC = 275$ m.

2. Resolve

Resolve AC :

$$\text{east component} = 275 \sin 120^\circ = 238.16 \dots$$

$$\text{north component} = 275 \cos 120^\circ = -137.5$$

3. Use bearings

Ship B is 150 m north of A , so from B to C :

$$\text{east} = 238.16 \dots, \quad \text{north} = -137.5 - 150 = -287.5$$

4. Use bearings

The bearing is therefore in the south-east quadrant:

$$180^\circ - \tan^{-1} \left(\frac{238.16 \dots}{287.5} \right) = 140.36 \dots^\circ$$

FINAL ANSWER

140° .

PAST PAPER QUESTION**Q#: 17****Marks: 5****TOPIC: Differentiation**

Jan 2021 P1H - Jan 2021 | P1H

17 A solid, **S**, is made from a hemisphere and a cylinder.

The centre of the circular face of the hemisphere and the centre of the top face of the cylinder are at the same point.

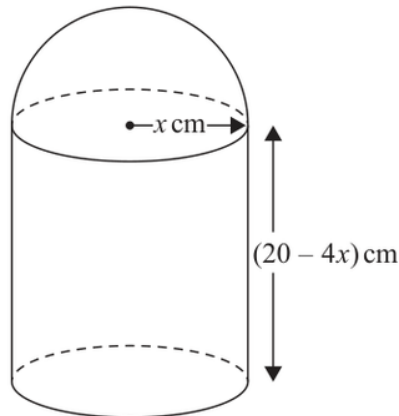


Diagram **NOT**
accurately drawn

The radius of the cylinder and the radius of the hemisphere are both x cm.

The height of the cylinder is $(20 - 4x)$ cm.

The volume of **S** is $V \text{ cm}^3$ where $V = \frac{1}{3} \pi y$

Find the maximum value of y .
Show clear algebraic working.

(Total for Question 17 is 5 marks)

PAST PAPER SOLUTION**Q#: 17****Marks: 5****TOPIC: Differentiation**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Volume of the solid is**

Volume of the solid is cylinder plus hemisphere:

$$V = \pi x^2(20 - 4x) + \frac{2}{3}\pi x^3$$

$$V = 20\pi x^2 - 4\pi x^3 + \frac{2}{3}\pi x^3$$

$$V = 20\pi x^2 - \frac{10}{3}\pi x^3$$

2. The question gives

The question gives

$$V = \frac{1}{3}\pi y$$

3. Differentiate

So

$$y = 60x^2 - 10x^3$$

4. Differentiate

Differentiate:

$$\frac{dy}{dx} = 120x - 30x^2$$

5. Differentiate

For a maximum,

$$120x - 30x^2 = 0$$

$$30x(4 - x) = 0$$

6. Differentiate

So the valid value is

$$x = 4$$

$$y = 60(4^2) - 10(4^3) = 960 - 640 = 320$$

FINAL ANSWER

The maximum value of y is 320.

EA

PAST PAPER QUESTION**Q#: 18****Marks: 3**TOPIC: **Surds**

Jan 2021 P1H - Jan 2021 | P1H

- 18** Given that $(8 - \sqrt{x})(5 + \sqrt{x}) = y\sqrt{x} + 21$ where x is a prime number and y is an integer, find the value of x and the value of y . Show each stage of your working clearly.

 $x = \dots\dots\dots$ $y = \dots\dots\dots$ **(Total for Question 18 is 3 marks)**

EA

PAST PAPER SOLUTION**Q#: 18****Marks: 3****TOPIC: Surds**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Simplify surd**

$$\begin{aligned}(8 - \sqrt{x})(5 + \sqrt{x}) &= 40 + 8\sqrt{x} - 5\sqrt{x} - x \\ &= 40 + 3\sqrt{x} - x\end{aligned}$$

2. Simplify surd

This is equal to

$$y\sqrt{x} + 21$$

3. Compare the rational parts

Compare the rational parts:

$$40 - x = 21$$

$$x = 19$$

4. Compare the surd coefficients

Compare the surd coefficients:

$$y = 3$$

FINAL ANSWER

$$x = 19, y = 3$$

PAST PAPER QUESTION**Q#: 19****Marks: 5****TOPIC: Simultaneous Equations**

Jan 2021 P1H - Jan 2021 | P1H

19 Solve the simultaneous equations

$$x^2 - 9y - x = 2y^2 - 12$$

$$x + 2y - 1 = 0$$

Show clear algebraic working.

(Total for Question 19 is 5 marks)

EA

PAST PAPER SOLUTION**Q#: 19****Marks: 5****TOPIC: Simultaneous Equations**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Solve simultaneous equations**

$$x^2 - 9y - x = 2y^2 - 12$$

$$x + 2y - 1 = 0$$

2. From the second equation

From the second equation:

$$x = 1 - 2y$$

3. Substitute into the first equation

Substitute into the first equation:

$$(1 - 2y)^2 - 9y - (1 - 2y) = 2y^2 - 12$$

$$4y^2 - 11y = 2y^2 - 12$$

$$2y^2 - 11y + 12 = 0$$

4. Factorise

Factorise:

$$(2y - 3)(y - 4) = 0$$

5. Solve simultaneous equations

So

$$y = \frac{3}{2} \quad \text{or} \quad y = 4$$

6. Solve simultaneous equationsIf $y = \frac{3}{2}$:

$$x = 1 - 2\left(\frac{3}{2}\right) = -2$$

7. Solve simultaneous equationsIf $y = 4$:

$$x = 1 - 8 = -7$$

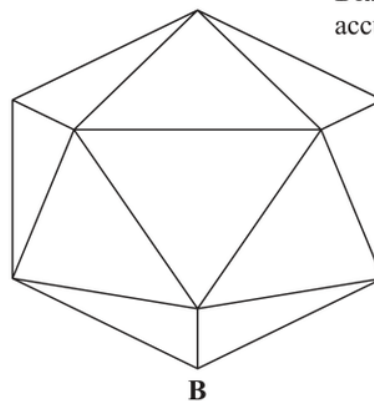
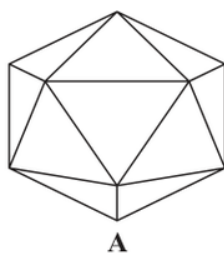
FINAL ANSWER

$$(x, y) = \left(-2, \frac{3}{2}\right) \text{ or } (-7, 4).$$

EA

PAST PAPER QUESTION**Q#: 20****Marks: 3****TOPIC: Area & Volume of Similar Shapes**

Jan 2021 P1H - Jan 2021 | P1H

20 **A** and **B** are two similar solids.Diagram **NOT**
accurately drawn**A** has a volume of 1836 cm^3 **B** has a volume of 4352 cm^3 **B** has a total surface area of 1120 cm^2 Work out the total surface area of **A**...... cm^2

(Total for Question 20 is 3 marks)

PAST PAPER SOLUTION**Q#: 20****Marks: 3****TOPIC: Area & Volume of Similar Shapes**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The volume ratio of solid**

The volume ratio of solid A to solid B is

$$1836 : 4352$$

2. Simplify

Simplify:

$$\frac{1836}{4352} = \frac{27}{64}$$

3. Split the ratio

So the linear ratio is

$$\sqrt[3]{27} : \sqrt[3]{64} = 3 : 4$$

4. Surface area ratio

Surface area ratio:

$$3^2 : 4^2 = 9 : 16$$

5. Solid B has surface areaSolid B has surface area 1120 cm^2 , so solid A has surface area

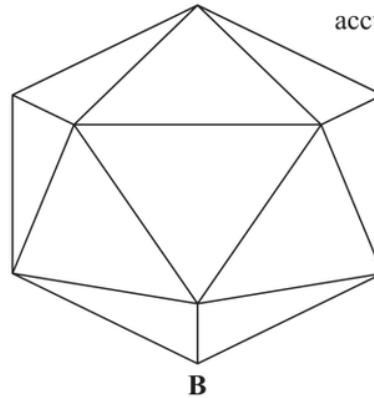
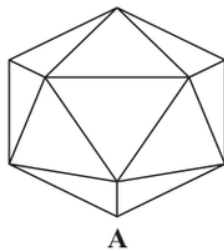
$$1120 \times \frac{9}{16} = 630$$

FINAL ANSWER

$$630 \text{ cm}^2$$

PAST PAPER QUESTION**Q#: 20****Marks: 3****TOPIC: Area & Volume of Similar Shapes**

Jan 2021 P1H - Jan 2021 | P1H

20 **A** and **B** are two similar solids.Diagram **NOT**
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PAST PAPER SOLUTION**Q#: 20****Marks: 3****TOPIC: Area & Volume of Similar Shapes**

Jan 2021 P1H - Jan 2021 | P1H

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FINAL ANSWER

$$630 \text{ cm}^2$$

PAST PAPER QUESTION**Q#: 21** **Marks: 4****TOPIC: Transformations of Graphs**

Jan 2021 P1H - Jan 2021 | P1H

21 A curve has equation $y = f(x)$ The coordinates of the minimum point on this curve are $(-9, 15)$

(a) Write down the coordinates of the minimum point on the curve with equation

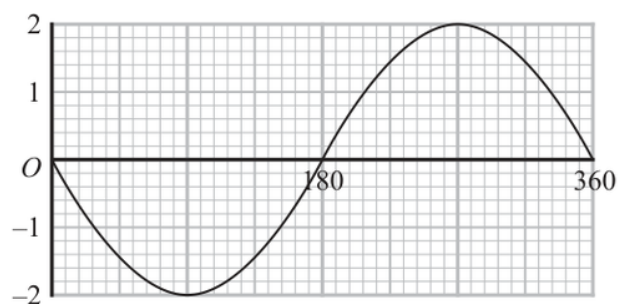
(i) $y = f(x + 3)$

(.....,)

(ii) $y = \frac{1}{3}f(x)$

(.....,)

(2)

The graph of $y = a \cos(x + b)^\circ$ for $0 \leq x \leq 360$ is drawn on the grid below.Given that $a > 0$ and that $0 < b < 360$ (b) find the value of a and the value of b . $a = \dots\dots\dots$ $b = \dots\dots\dots$

(2)

(Total for Question 21 is 4 marks)

PAST PAPER SOLUTION**Q#: 21****Marks: 4****TOPIC: Transformations of Graphs**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The minimum point of is**The minimum point of $y = f(x)$ is $(-9, 15)$.**2. Calculate value**

For

$$y = f(x + 3)$$

3. the graph moves 3 units

the graph moves 3 units left:

$$(-9, 15) \rightarrow (-12, 15)$$

4. Evaluate fraction

For

$$y = \frac{1}{3}f(x)$$

5. the coordinate is divided bythe y -coordinate is divided by 3:

$$(-9, 15) \rightarrow (-9, 5)$$

6. Use trigonometryFor part (b), the cosine graph has maximum 2 and minimum -2 , so

$$a = 2$$

7. The graph has a minimumThe graph has a minimum at $x = 90^\circ$. For $a \cos(x + b)^\circ$, a minimum occurs when

$$x + b = 180$$

8. Calculate value

So

$$90 + b = 180$$

$$b = 90$$

FINAL ANSWER

$(-12, 15), (-9, 5), a = 2, b = 90$

EA

PAST PAPER QUESTION**Q#: 21****Marks: 4****TOPIC: Transformations of Graphs**

Jan 2021 P1H - Jan 2021 | P1H

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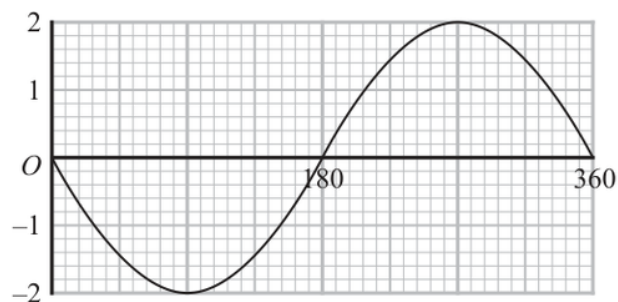
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(2)

(Total for Question 21 is 4 marks)

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Jan 2021 P1H - Jan 2021 | P1H

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8. Calculate value

So

$$90 + b = 180$$

$$b = 90$$

FINAL ANSWER $(-12, 15), (-9, 5), a = 2, b = 90$

EA

PAST PAPER QUESTION**Q#: 22****Marks: 3****TOPIC: Functions**

Jan 2021 P1H - Jan 2021 | P1H

22 The function f is such that $f(x) = x^2 - 8x + 5$ where $x \leq 4$

Express the inverse function f^{-1} in the form $f^{-1}(x) = \dots$

$f^{-1}(x) = \dots\dots\dots$

(Total for Question 22 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 22****Marks: 3****TOPIC: Functions**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate value**

$$f(x) = x^2 - 8x + 5$$

$$f(x) = (x - 4)^2 - 11$$

2. Let

Let

$$y = (x - 4)^2 - 11$$

$$y + 11 = (x - 4)^2$$

3. Find inverse functionSince the domain is $x < 4$, take the negative square root:

$$x - 4 = -\sqrt{y + 11}$$

$$x = 4 - \sqrt{y + 11}$$

FINAL ANSWER

$$f^{-1}(x) = 4 - \sqrt{x + 11}$$

PAST PAPER QUESTION**Q#: 22****Marks: 3****TOPIC: Functions**

Jan 2021 P1H - Jan 2021 | P1H

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$f^{-1}(x) = \dots\dots\dots$

(Total for Question 22 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 22****Marks: 3****TOPIC: Functions**

Jan 2021 P1H - Jan 2021 | P1H

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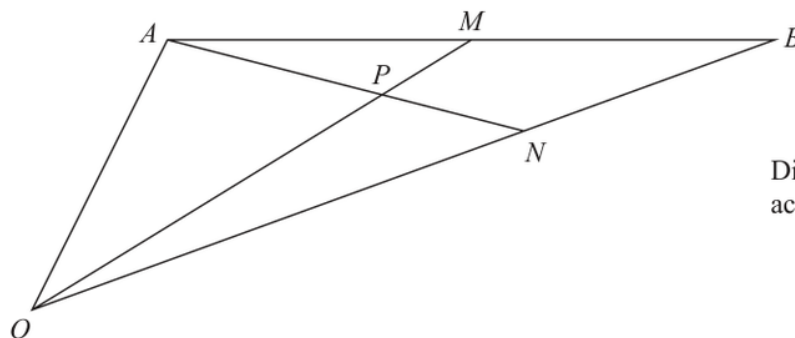
$$x = 4 - \sqrt{y + 11}$$

FINAL ANSWER

$$f^{-1}(x) = 4 - \sqrt{x + 11}$$

PAST PAPER QUESTION**Q#: 23** **Marks: 6**TOPIC: **Vectors**

Jan 2021 P1H - Jan 2021 | P1H

23 OAB is a triangle.Diagram **NOT**
accurately drawn

$$\vec{OA} = 2\mathbf{a} \text{ and } \vec{OB} = 2\mathbf{b}$$

 M is the midpoint of AB . N is the point on OB such that $ON:NB = 2:1$ P is the point on AN such that OPM is a straight line.Use a vector method to find $OP:PM$

Show your working clearly.

(Total for Question 23 is 6 marks)

PAST PAPER SOLUTION**Q#: 23****Marks: 6**TOPIC: **Vectors**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use vectors**

$$\vec{OA} = 2\mathbf{a}, \quad \vec{OB} = 2\mathbf{b}$$

2. Find the midpointSince M is the midpoint of AB ,

$$\vec{OM} = \frac{2\mathbf{a} + 2\mathbf{b}}{2}$$

$$\vec{OM} = \mathbf{a} + \mathbf{b}$$

3. Use vectorsSince $ON : NB = 2 : 1$,

$$\vec{ON} = \frac{2}{3}\vec{OB}$$

$$\vec{ON} = \frac{2}{3}(2\mathbf{b}) = \frac{4}{3}\mathbf{b}$$

4. Use vectorsPoint P lies on OM , so

$$\vec{OP} = t(\mathbf{a} + \mathbf{b})$$

5. Use vectorsPoint P also lies on AN , so

$$\vec{OP} = 2\mathbf{a} + s\left(\frac{4}{3}\mathbf{b} - 2\mathbf{a}\right)$$

$$\vec{OP} = (2 - 2s)\mathbf{a} + \frac{4s}{3}\mathbf{b}$$

6. Use vectorsEquate coefficients of $\mathbf{a}$ and $\mathbf{b}$:

$$t = 2 - 2s$$

$$t = \frac{4s}{3}$$

7. Use vectors

So

$$\frac{4s}{3} = 2 - 2s$$

$$4s = 6 - 6s$$

$$10s = 6$$

$$s = \frac{3}{5}$$

8. Use vectors

Then

$$t = \frac{4}{5}$$

9. Use vectors

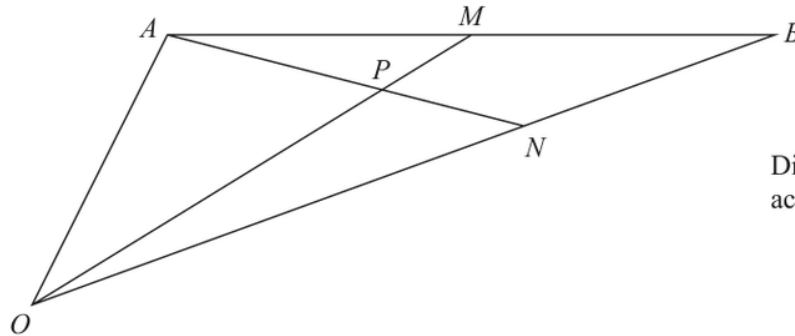
So P is $\frac{4}{5}$ of the way from O to M .

FINAL ANSWER

$$OP : PM = 4 : 1$$

PAST PAPER QUESTION**Q#: 23****Marks: 6**TOPIC: **Vectors**

Jan 2021 P1H - Jan 2021 | P1H

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Show your working clearly.

(Total for Question 23 is 6 marks)

PAST PAPER SOLUTION

Q#: 23

Marks: 6

TOPIC: **Vectors**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Use vectors

$$\vec{OA} = 2\mathbf{a}, \quad \vec{OB} = 2\mathbf{b}$$

2. Find the midpointSince M is the midpoint of AB ,

$$\vec{OM} = \frac{2\mathbf{a} + 2\mathbf{b}}{2}$$

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8. Use vectors

Then

$$t = \frac{4}{5}$$

9. Use vectors

So P is $\frac{4}{5}$ of the way from O to M .

FINAL ANSWER

$$OP : PM = 4 : 1$$

EA

PAST PAPER QUESTION**Q#: 24****Marks: 4****TOPIC: Sequences**

Jan 2021 P1H - Jan 2021 | P1H

24 An arithmetic series has first term a and common difference d .

The sum of the first $2n$ terms of the series is four times the sum of the first n terms of the series.

Find an expression for a in terms of d .
Show your working clearly.

 $a = \dots\dots\dots$

(Total for Question 24 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 24****Marks: 4****TOPIC: Sequences**

Jan 2021 P1H - Jan 2021 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Evaluate fraction**

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

$$S_{2n} = n(2a + (2n-1)d)$$

2. Given

Given

$$S_{2n} = 4S_n$$

$$n(2a + (2n-1)d) = 4 \cdot \frac{n}{2}(2a + (n-1)d)$$

3. Divide byDivide by n :

$$2a + (2n-1)d = 2(2a + (n-1)d)$$

$$2a + 2nd - d = 4a + 2nd - 2d$$

$$d = 2a$$

$$a = \frac{d}{2}$$

FINAL ANSWER

$$a = \frac{d}{2}$$

PAST PAPER QUESTION**Q#: 24****Marks: 4****TOPIC: Sequences**

Jan 2021 P1H - Jan 2021 | P1H

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(Total for Question 24 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 24****Marks: 4**TOPIC: **Sequences**

Jan 2021 P1H - Jan 2021 | P1H

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FINAL ANSWER

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